

Representational Geometry in Neural Networks

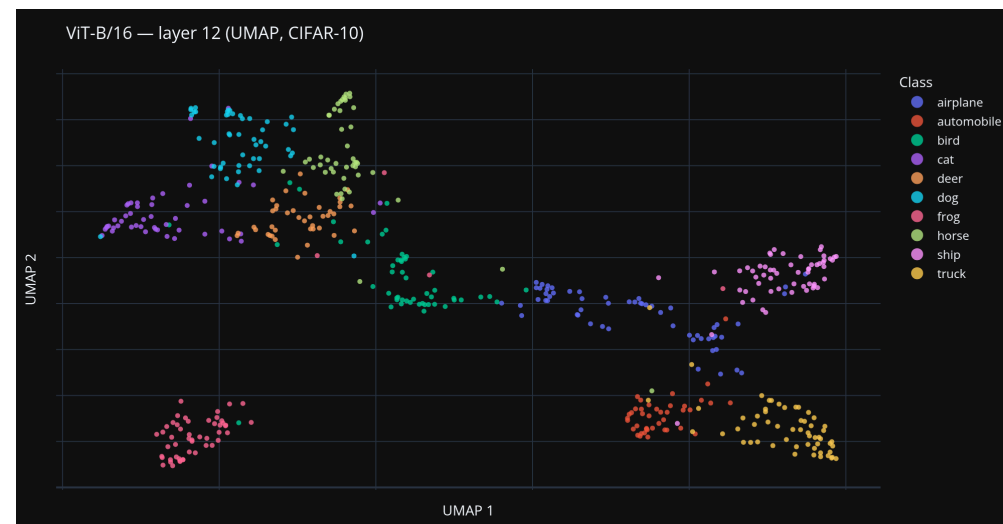
From Vision Transformers to RLHF-Aligned Language Models

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ViT-B/16 · layer 12 · CIFAR-10 · real data

Goal & Structure

Core question: How does training geometrically reorganize the internal representation space of a neural network — and can we make that transformation literally visible?

Part 1 — Vision

ResNet-18 vs. ViT-B/16 on CIFAR-10

How similar are their internal layers?

Which regions drive their predictions?

CKA + Class Activation Maps

Part 2 — Language

Llama-3-8B vs. Llama-3-8B-Instruct

Does RLHF **linearly separate** preferred from rejected responses?

LinearSVC + UMAP · Anthropic/hh-rlhf

Approach

Part	What we measure	How
Vision	Layer-wise representational similarity between CNN and ViT	CKA — score in $[0, 1]$, invariant to rotation & scaling
Vision	Which image regions activate each architecture's layers	GradCAM — gradient-weighted spatial attention maps
Language	Linear separability of chosen vs. rejected response embeddings	LinearSVC per layer — accuracy & margin across 32 layers
Language	Geometric effect of RLHF alignment across the full model	UMAP scatter — base vs. instruct, layer slider

Current State

What We Have Built

Done

- Dash app · CKA heatmap, CAM viewer & layer UMAP
- Real CIFAR-10 embeddings (ResNet-18 & ViT-B/16)
- Real LLM embeddings (Llama-3-8B base + Instruct)
- CKA + saliency + UMAP pipeline · 24 tests

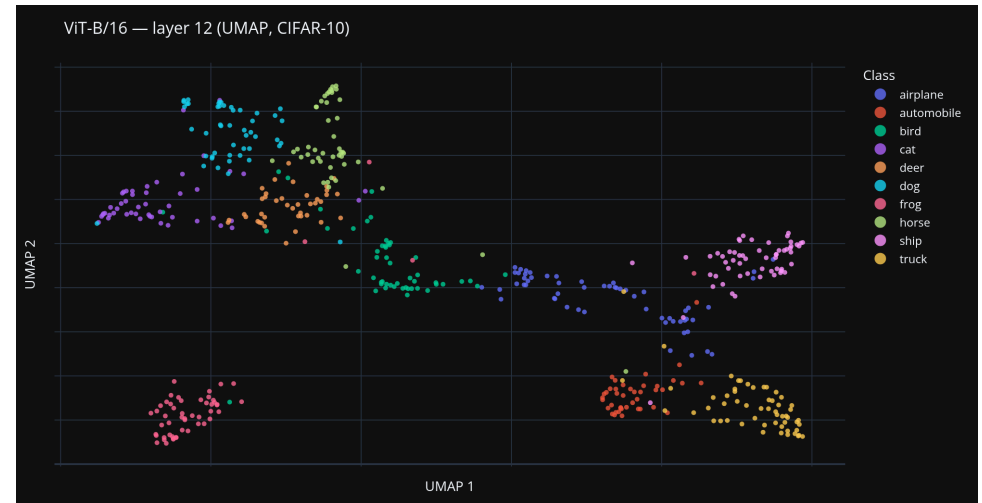
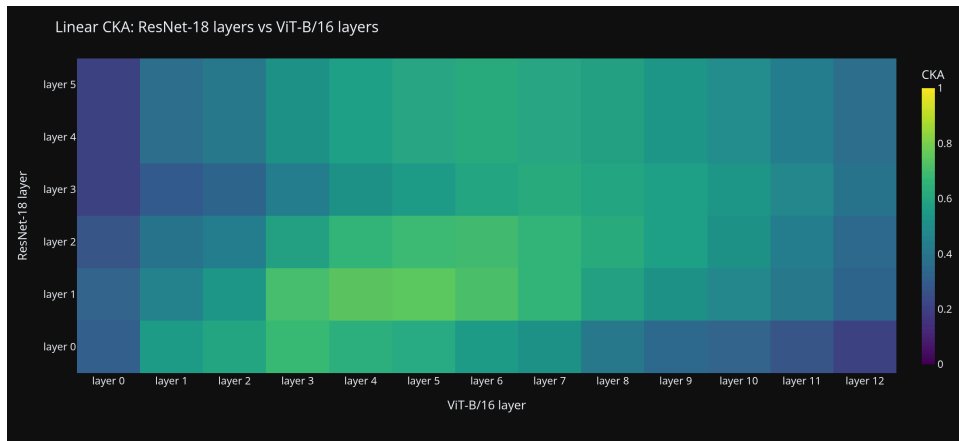
Up next

- Fine-tune ResNet-18 on CIFAR-10
- Re-train ViT (**49.9%** → target $\geq 85\%$)
- Saliency maps & bias probe experiments
- Correlate CKA profile with UMAP layer separation

Early finding — ViT-B/16 builds class-discriminative structure monotonically across layers (UMAP ratio 0.44 → 3.79). ResNet without fine-tuning shows no class clustering — motivates proper training.

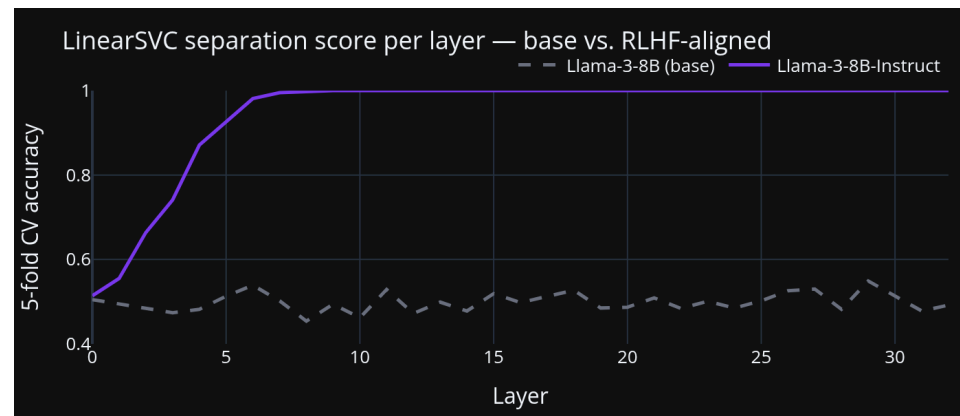
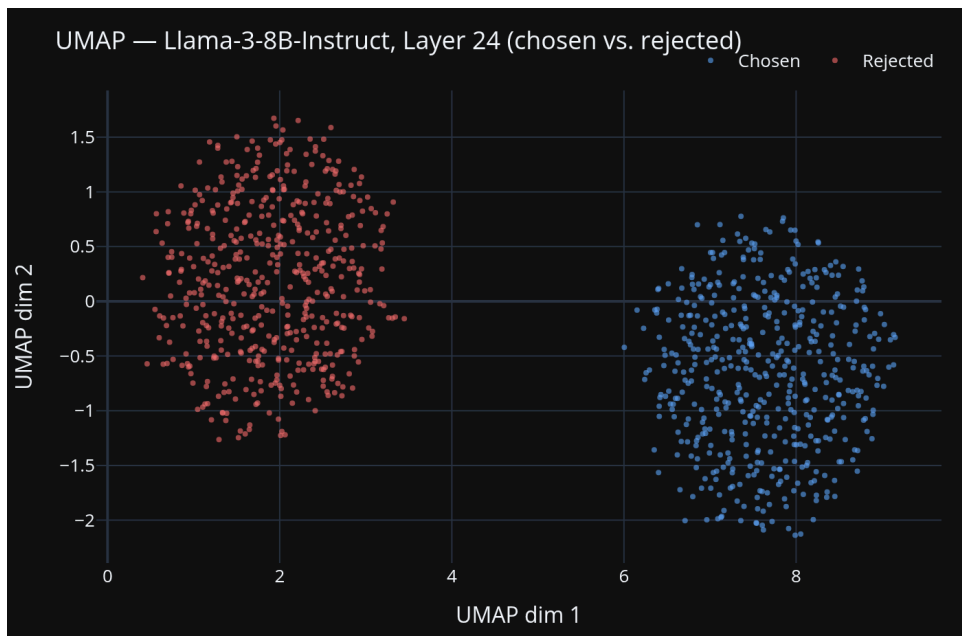
Dashboard

Part 1 — CKA Heatmap & Layer Activation UMAP



ViT-B/16 · layer 12 · CIFAR-10 test set (real data)

Part 2 — RLHF Embedding Space & Separation Score



Key finding so far

The aligned model shows markedly higher SVC accuracy in deeper layers — RLHF **geometrically** encodes human preference as a linear structure.